

ARDL Output Explanation - M.L. in Finance

AutoRegressive Distributed Lag Model (ARDL) :

An AutoRegressive Distributed Lag (ARDL) model is an econometric tool for analyzing time series data, modeling how a dependent variable depends on its own past values (AutoRegressive part) and past values of independent variables (Distributed Lag part). It can even handle variables with different integration orders, whether stationary or nonstationary, but for this project I chose to make the data stationary by taking first differences.

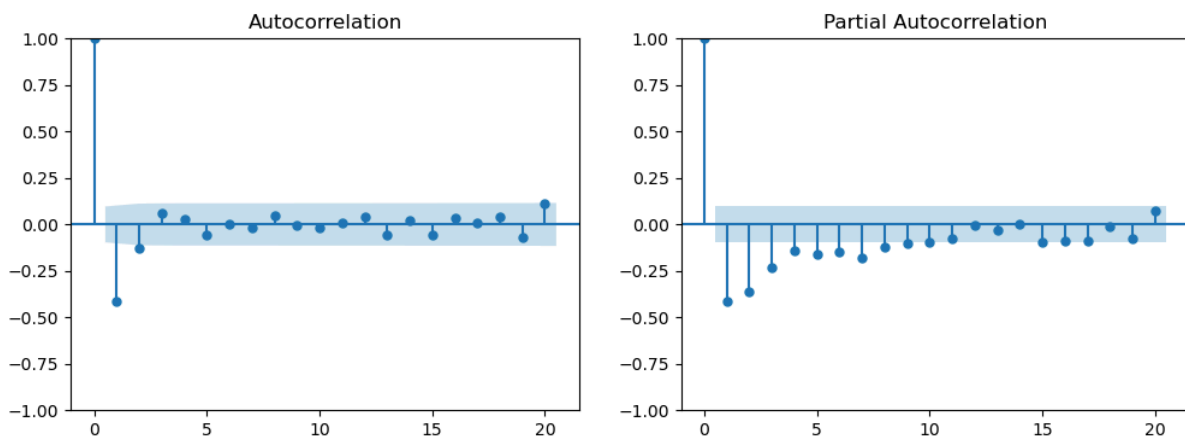
An ARDL(p,q) model for which Y_t is equal to past values of Y_t and current and past values of some regressor Z_t plus error term U_t :

$$y_t = \rho_0 + \rho_1 y_{t-1} + \dots + \rho_p y_{t-p} + \alpha'_1 z_t + \dots + \alpha'_{q+1} z_{t-q} + u_t$$

ARDL is a simple model that uses classical OLS regression to estimate its coefficients. It is less complex than Machine Learning techniques, and it will struggle more to find long-term relationships. However, on the upside, it doesn't have this 'black box' that Machine Learning models have. It is a fully transparent and interpretable model.

Interpretation of the ARDL model results :

1) Autocorrelation and Partial Autocorrelation :



ACF and PACF of the US T-note 10-year yield. It shows us how its lagged values are correlated between them and the current value. Only spikes that cross the blue interval are considered relevant.

2) Augmented Dickey-Fuller (ADF) Test :

US T-Note 10Y Yield	Stationary	ADF Statistic: -8.69, p-value: 0.0
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US GDP	Stationary	ADF Statistic: -4.627, p-value: 0.0
US T-Note 5Y Yield	Stationary	ADF Statistic: -18.283, p-value: 0.0
Interest Rates	Stationary	ADF Statistic: -4.486, p-value: 0.0
US T-Note 2Y Yield	Stationary	ADF Statistic: -4.874, p-value: 0.0
XAU/USD	Stationary	ADF Statistic: -3.803, p-value: 0.003
US T-Bond 30Y Yield	Stationary	ADF Statistic: -15.285, p-value: 0.0
OIL WTI	Stationary	ADF Statistic: -16.624, p-value: 0.0
US Unemployment	Stationary	ADF Statistic: -8.201, p-value: 0.0
USD/EUR	Stationary	ADF Statistic: -8.72, p-value: 0.0
USD Index	Stationary	ADF Statistic: -18.773, p-value: 0.0
VIX	Stationary	ADF Statistic: -9.623, p-value: 0.0

With time series forecasting, it is generally better to perform the regression on stationary data; in other terms, after removing the trend and seasonality. For this process, I used the first-order integration of the data, $I(1)$, we get it by differencing the current value with the first lagged value. The ADF test tests whether the data is stationary.

3) ARDL Forecasting Metrics :

MSE = 0.084

R² = 0.32

Adjusted R² = 0.239

The ARDL MSE score is lower than the one obtained with the Random Forest model, meaning it got a closer prediction to the actual values. However, the R² and adjusted R² scores are less than 5%, which means to a really poor goodness-of-fit of the dataset. We can expect that the prediction will look right, but it is mostly due to luck/randomness.

4) ARDL Model Results :

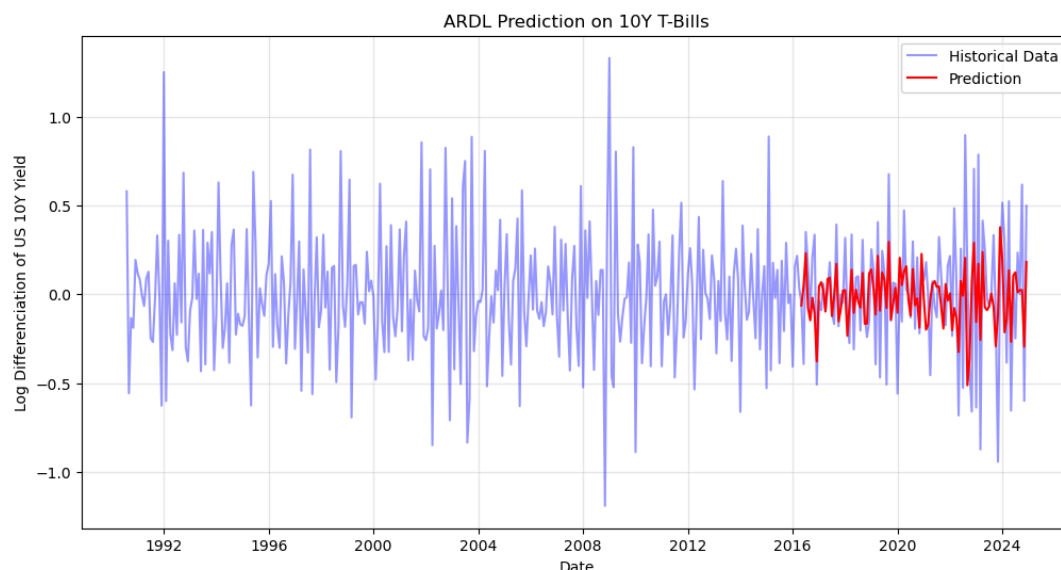
ARDL Model Results

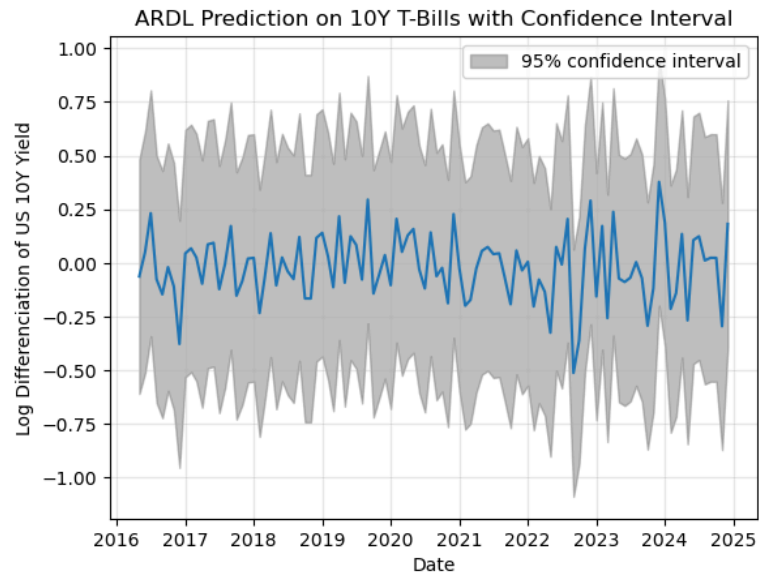
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Dependant Variable :   US T-Note 10Y Yield | No. Observations :           309
Model : ARDL(2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1) | Log Likelihood :       -43.483
Method :               Conditional MLE | S.D. of innovations :       0.279
AIC :                  116.966
BIC :                  172.869
HQIC :                 139.321
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	coefficient	std err	z	P> z	[0.025	0.975]
const	-0.0120	0.025	-0.476	0.634	-0.062	0.038
US T-Note 10Y Yield.L1	-0.2470	0.076	-3.254	0.001	-0.396	-0.098
US T-Note 10Y Yield.L2	-0.2534	0.058	-4.387	0.000	-0.367	-0.140
USD/EUR.L1	1.0048	0.645	1.558	0.120	-0.265	2.274
US Unemployment.L1	0.0527	0.061	0.858	0.392	-0.068	0.173
US GDP.L1	1.664e-05	0.000	0.081	0.936	-0.000	0.000
XAU/USD.L1	-3.409e-05	0.000	-0.082	0.934	-0.001	0.001
USD Index.L1	-0.0072	0.011	-0.673	0.501	-0.028	0.014
OIL WTI.L1	0.0018	0.004	0.468	0.640	-0.006	0.009
VIX.L1	-0.0072	0.004	-1.786	0.075	-0.015	0.001
Interest Rates.L1	-0.0746	0.106	-0.703	0.482	-0.283	0.134
US T-Bond 30Y Yield.L1	0.0192	0.010	1.975	0.049	6.61e-05	0.038
US T-Note 5Y Yield.L1	0.1071	0.045	2.365	0.019	0.018	0.196
US T-Note 2Y Yield.L1	-0.1379	0.087	-1.584	0.114	-0.309	0.033

We notice that the only features having a P-value lower than 0.05 are: the US T-Note 10Y Yield at lag 1 and 2, the US T-Note 5Y Yield at lag 1, and the US T-Bond 30Y Yield at lag 1. Meaning that all of the other features are not statistically significant at a confidence level of 95%.

5) Plot of the ARDL prediction VS. Actual Data :





From a visual point of view the model prediction doesn't seem that off. However, I assume the prediction is mostly noise / non relevant by the P-Values and metrics (R2) we have seen. The confidence interval of our model prediction is quite large; the model is not accurate.